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1) Technique : Negative transformation

- Negative transformation converts bright pixels into dark pixels

$$S = L - 1 - r$$

It reduces excessive brightness & makes hidden details visible

Result: Better visibility & balanced brightness.

2) Dark Image Lack details due to log transformation increase the brightness of dark pixels.

Formula =>

$$S = C\log(1+r)$$

It enhances low-intensity regions & reveals hidden details.

Result: Dark areas becomes brighter & clearer.

3) Histogram Equalization spreads pixel intensities over full range

It increases difference between dark & bright regions

Result: Improved contrast & better image quality.

4) Low contrast Image requires Uniform Intensity distribution

It has to use Histogram equalization -

~~It~~ redistributes pixel intensities uniformly

Result: Better contrast & Improved visibility.

5) Image shows uneven brightness

Technique - Adaptive Histogram Equalization (CLAHE)

CLAHE enhances small regions of image seperately & It corrects the Uneven lighting without overenhancing the image.

Result : Uniform brightness & clearer details.

6) Noisy Image contains random intensity variations

Technique - Median Filter.

-> Median filter replaces each pixel with the median of its neighbours.

-> It removes salt & pepper noise while preserving edges.

Result : Reduced noise & smooth image.

7) Noise Reduction without loosing details.

Technique : Bilateral Filter.

- Bilateral Filter smooths the image while preserving edges. It considers both pixel distance & intensity differences.

Result : Noise is reduced & image details remain sharp

8) Image appears blurred after smoothing

Technique : Laplacian sharpening

- Laplacian sharpening enhances image edges. It restores details lost during smoothing

Result : Sharper Image with clearer edges.

17) 9) Fine details in an image are not clearly visible.

Technique - High Pass filter.

- A High pass filter enhances fine details & edges. It emphasizes high frequency components while reducing low freq components.

Result: Sharper image with clearer details.

10) A Noisy image must be pre processed before further analysis

Technique - Median filter.

- The median filter removes noise by replacing each pixel with the median of neighboring pixels. It preserves edges while reducing noise.

Result: Cleaner image suitable for further processing

11) An image contains High frequency Noise

Technique - Low Pass filter

-> It removes high freq noise. it allows low freq components to pass & block high frequencies.

Result: Smoother image with reduced noise.

12) Preserve low-freq components while removing noise.

Technique - Ideal low pass filter [ILPF]

-> It passes low freq components & block high freq noise. It smooths the image while keeping overall brightness.

Result: Reduced Noise & Preserved Image structure.

F 13) Image requires edge enhancement in the frequency domain.

Technique : High Pass filter.

→ A High pass filter highlights edges by passing high freq components
It suppresses smooth regions

Result : Enhanced edges & sharper images.

14) A smooth version of image is required

Technique - Low-pass filter

A low pass filter removes high freq components, It smooths the image & reduce noise.

Result : Blur free noise reduction & smoother appearance.

15) An Object is clearly brighter than the background.

Technique : Thresholding

Pixels above the threshold become the object others become background & seperates the Objects using an intensity threshold.

Result : Clear object Segmentation

16) Seperate foreground & background based on Intensity

Technique - Global Thresholding

- A single threshold value divides image into foreground & background.
Pixels are classified based on their intensity.

Result : Simple & effective segmentation.

17) Image requires identification of object edges.

Technique: Canny Edge detection

→ Canny detects edges using gradient calculation & non-max supression. It produces thin & accurate edges.

Result: Clear & continous edge detection.

18) Adjacent pixels have similar intensity values.

Technique - Region growing

- Region growing groups neighbouring pixels with similar intensity. It starts from a seed pt & expands the region.

Result: similar regions are segmented accurately.

19) Detected edges appear broken

Technique - Morphological closing

→ Morphological closing connects broken edges using dilation followed by erosion. It fills small gaps in edges.

Result: Continous & connected boundaries

20) Extract object outlines from an image.

Technique: Boundary Extraction.

→ Boundary extraction identifies the outerbody of objects. It's commonly performed using morphological operations or edge detection.

Result: Clear object outlines for analysis.